Historical source

Encyclopaedia Britannica scan, source row 115. Parse 3.5 extracted the text. Super 3.5 VL corrected it from an enlarged crop of each block. It can still contain recognition errors. Page headers, footers, and page numbers are removed.

BLEACHING (E0003)

Continues an entry from an earlier page.

Vegetable time, and repeated proofs of the good quality of the Substances. fluid. At the end of eight hours distillation, I drew

of one-fourth of the contents of the veffly a bound fourth two hours afterwards; a third fourth after ten hours and a half of eleven hours; and the reffi after twelve hours affiliation, who formed the conditions.

When the liquor is entirely worn off from the veffel, it must not be immediately filled with water, or at least to the height of five or its inches above the return of the leader tube, otherwise the gas, which continues to escape from the difficulty veffle then affords no refinance, might attack the pneumatic veffel inficit.

250. "The fire must be taken from beneath the retort as soon as the distillation is finished, not only to prevent the effect of the gaseous vapours, which will continue slowly to escape, from acting on the sides of the tub, but likewise to dispose the retorts or bodies to receive a quantity of warm water, which is to be poured in up to the neck. There is no reason to fear an excess of quantity, and the hotter the vessels are the better. It is essential, however, that it be not poured in cold, for fear of breaking the glass. The adopter is then to be unloosed from the neck of the leaden tube, if the operator chooses; and in order that no vapour may escape into the workshop, a bit of lute or a cork may be applied to the beak of the adopter. The sand bath easily permits the retort to be raised and returned again to its place, as well as the application of the lute or flopper to the neck of the adopter, this last being raised with one hand while the cork is put in with the other.

Nevertheless, as the lutes which connect the adopter with the retort are somewhat loosened towards the end of the operation, it would be more prudent to leave every thing in its place, for fear of deranging the lutes. This danger is greater when the adopter is of lead, because the great length of this additional piece tends to force the luting fill more on that account. If it be required to proceed immediately to a new distillation, the retort or bottle with its capsule or pan must be immediately taken from the furnace, and another substituted in its place ready prepared during the former distillation. This necessarily requires a double set of vessels.

211. When the distilling vessel is cold, or nearly so, the whole of its contents must be shaken, by holding this vessel by the neck with one hand, and applying the other to its bottom. The stopper must then be taken out, and the vessel speedily inverted, shaking the residue to facilitate its escape. In this last situation the retort is to be held by its neck with one hand, and its side gently resting against the other. The vessels into which the water and residual matter of the retorts are poured, should rather be of stoneware, pottery, or lead, than of wood, unless the last be oil vessels, which are less subject to dry in the part above the fluid. If this circumstance be not attended to, there will be danger of losing great part of the contents.

It is most convenient to disengage the retorts or bodies while they are still warm, which continues to be the case the next morning after distillation, in con- sequence of the heat of the sand bath. If they be left to cool entirely, the sulphate of soda will crystallize, and it will be necessary to dissolve in hot water such

larger portions as cannot pass through the neck. But Vegetable this inconvenience is not likely to happen, unless the quantity of water left added be too small, and the re-

this inconvenience is not likely to happen, unless the quantity of water left added be too small, and the re- fiduies have been left undiluted for several days. The same observation is applicable to that kind of incru- tation which is formed by the muriat, if not properly pulverised, dried, or mixed ; this cannot be separated from the bottom of the retort, but by means of hot water poured at different succelive times. It is like- wise essential to leave no crust or deposition of muriat, or other matter, in the vessels which are emptied, un- less the fame be moveable, in which case the rirk is left. But if the urgency of business should then re- quire that the fame vessels be used without entirely clearing them, it will be necessary to range this resid- ual matter on one side, where it will be less exposed to the heat, and will afford a greater degree of facility for the nitric acid to act upon it.

In order that the vapour which exhales from the distilling vessels may not prove inconvenient, it is necessary to pour in a small quantity of alkaline lixivium in the first place, which instantly destroys the smell. This may be done immediately after the end of the distillation, and the weak alkaline solution may supply the place of the water used for diluting the residues. At the instant of pouring this lixivial water, a strong effervescence takes place; for which reason it is proper to pour it in by several successive portions, waiting a little between each time.

252. The oxygenated liquor prepared in this way is very effectual in bleaching the fluffs which are immersed in it, but it at the same time polishes an odour so foetid as to render its use unpleasant and even injurious to the workmen employed in the process, should any of it escape. "I have witnessed, (says O'Reilly) in an extensive manufactory near Paris, the cruel sufferings experienced by the wretched workmen, from the fuliginating vapours; I have seen them rolling on the ground in the excess of agony. Frequently even severe disorders are the consequence of the first effects produced by the oxygenated liquor." De Charme gives a long account of the sufferings which he underwent in his course of experiments with this acid. It produces symptoms completely resembling those of a violent cold, but which go off in a day or two after having defiled from the use of the liquor.

This inconvenience may be avoided in two ways;

This inconvenience may be avoided in two ways; 1st, By rendering the vessels for immersion so tight as to prevent the possibility of the escape of the gas, 2dly, By dissolving in the water some substance

2dly. By dissolving in the water some substance which has the property of so far neutralizing the acid as to correct its odour without destroying its bleaching quality to any considerable degree.

253. Mr Rupp of Manchester contrived a tub for immersion, which is admirably adapted to answer the first purpose. Its construction is simple, and not ex- pensive.

It would, therefore, be desirable to have an apparatus for the use of the pure oxygenated muriatic acid simply dissolved in water, which is at once the cheapest and best vehicle for it. This apparatus must be simple in its construction, and obtainable at a moderate expense; it must confine the liquor in such a manner as to pre- vent the escape of the oxygenated muriatic acid gas, which is not only a loss of power, but also an inconve- nience.

Source record

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Running header 6goa BLEACHING. Part II.